

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6.	(a)			Dark lines [or equivalent] (1) Bright background [accept rainbow background or equivalent] (1) Dark lines in same positions [or for same wavelengths] as [some of the] bright lines in emission spectrum (1)	3			3		
	(b)	(i)	I	Electron given energy or pumped' (1) [Electron having fallen from P to U] increases population of U or brings about population inversion (1)	2			2		
			II	Electron falls quickly or spontaneously [from L to level G] (1) So depopulating level L or making pop inversion easier to establish (1)	2			2		
		(ii)		Photon energy = $(1.43 - 0.26)$ [eV] [= 1.17 eV] or by impl (1) Conversion to J even if incorrect energy gap [= 1.87×10^{-19} J or by implication] (1) Correct use of $\lambda = \frac{hc}{E}$ or $\lambda = \frac{c}{f}$ and $f = \frac{E}{h}$ ecf on E (1) 1 060 n[m] no ecf and infra-red (1)		4		4	3	
	(c)			A maximum of (3) marks by addition from... • Laser pointer (arguably) too low-powered to be dangerous • Lecturer should be trusted not to shine lasers at students (or strongly reflecting surfaces) or lasers not viewed directly • Might stop sight of bright spot even when pointer used correctly • Any good physics involving polarisation • Will make field of view dimmer so note-taking difficult or opening pupils of eyes, making laser beam <i>more</i> dangerous • Any other valid point • Conclusion argued using more than one of the above			3	3		
				Question 6 total	7	4	3	14	3	0